Role of Artificial Intelligence in Early Diagnosis of Cancer

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Abstract: *Cancer is oncogenic transformation of cells stimulated by carcinogens which stops contact inhibition that leads to tumors (benign or malign). As per World Health Organization (WHO), 10 million people died in 2020 due to cancer, most prominently by lung cancer, colon cancer, liver cancer, stomach cancer and breast cancer. Nearly, 50% cancer patients die due to late diagnosis. Current diagnostic techniques include biopsy and histopathology of tissues, radiography, computed tomography (CT), and magnetic resonance imaging (MRI). Some treatment techniques include chemotherapy, hormone therapy, and surgery. Although medical sector have these techniques, cancer at higher stages is still incurable, therefore late detection of cancer is fatal. With the onset of Industry 4.0, the era of Artificial Intelligence (AI) has been established. AI can be used to speed up the process of medical diagnosis, for example, convolutional neural network (CNN) to detect cancer based on medical report of the patient (X-ray, CT scan or MRI scan.*

Keywords: Artificial Intelligence, Cancer Diagnostics, Medical Imaging Techniques, Neural Networks, Industry 4.0.

1. Introduction

Although the advancement in technology led to the enhancement of treatment of a variety of diseases including cancer, still cancer is fatal & incurable at higher stages. In most of the cases, late diagnosis of cancer increases the risk of survival of patient. Therefore, cancer diagnostic technique plays a crucial role in treatment & survival of patients. Conventional diagnostic techniques such as Medical Imaging including Radiography, Mammography, Computer-Aided Detection (CAD), etc. are time consuming as well as less accurate, due to lot of dependences & constraints. Traditional radiography techniques depends majorly on '*semantic features'* (qualitative & quantitative analysis of phenotype & anatomy of cancer) such as cellular composition, tumour density, shape & size, etc. These techniques are crucial for detection of cancer therefore there is a huge need to enhance them.

With the onset of Industry 4.0, AI came into picture. As AI has a lot of potential to handle big data & to classify objects based on algorithms, therefore it can be implemented for disease diagnostics for faster results with better accuracy. Due to its overwhelming potential, AI has developed its roots in many sectors including business & market, agriculture, food, education, & even healthcare.

In this review, we have examined several ways for cancer diagnostics based on traditional biological methods as well as through AI algorithms which can provide better results in lesser time.

2. Cancer Statistics

Cancer is not a new disease but it was always there, as identified by Hippocrates, '*The Father of Medicine*'. Cancer is the abnormal transformation of

cells that exponentially grow & spread to other body parts through '*Metastasizing*'. Cancer is very deadly and is responsible for a large number of deaths globally, accounting for 10 million deaths in 2020, prominently by lung cancer (1.8 million deaths), colon & rectum cancer (916000 deaths), liver cancer (830000 deaths), stomach cancer (769000 deaths) and breast cancer (685000 deaths) [1,2], which can be seen in **Figure 1**, **2** and **3**, respectively. Due to lack of efficient & faster diagnostic techniques, the cancer crosses the curable stages & becomes fatal. It is estimated that by 2050, there will be around 35 million new cancer cases, i.e. 77% more than 20 million cases in 2022. Due to this impending danger, urgent measures have to be taken in order to save the people from cancer. One of the approaches is AI i.e. developing at a greater rate. AI algorithms can be developed that can detect cancer at early stages which can help to fasten the treatment process.

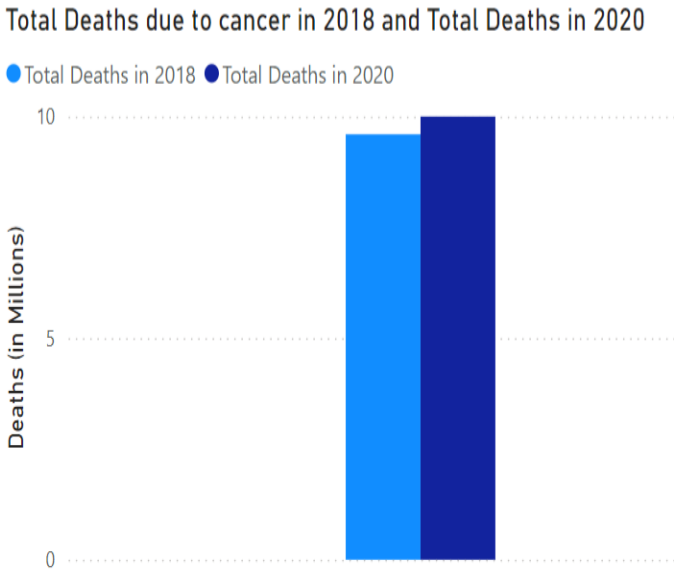

Figure 1. Total deaths due to cancer in 2018 and 2022 globally (in millions)
*source: World Health Organization (WHO) data

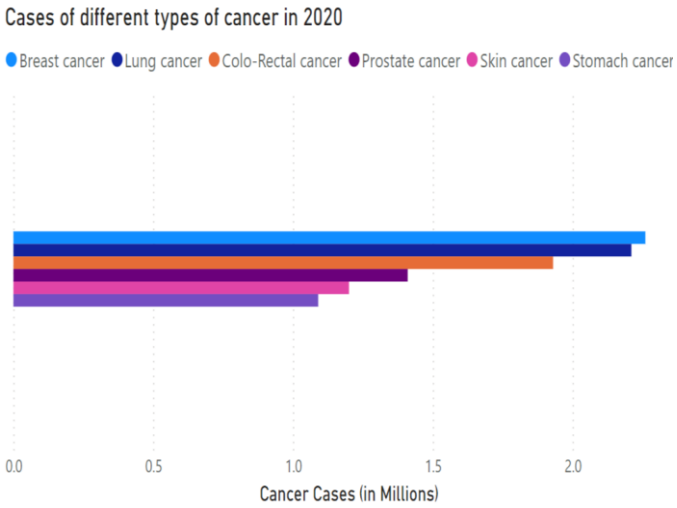

Figure 2. Total cases of different types of cancer globally in 2020
*source: World Health Organization (WHO) data

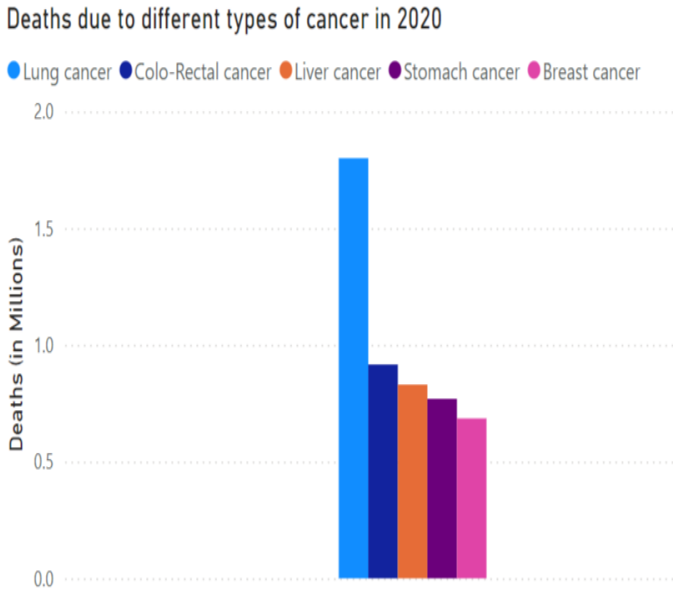

Figure 3. Total deaths due to different types of cancer globally in 2020
*source: World Health Organization (WHO) data

3. Birth of AI

John McCarthy coined the term 'Artificial Intelligence' or AI in Dartmouth conference in 1956. He explained that "AI is the science and engineering of making intelligent machines". Over the decades, the definition of AI keeps on changing, but the motive remains same, i.e. to automate & replicate the tasks that humans do to assist humanity by providing quicker solutions, suggestions, predictions & decisions over data. AI is currently a developing field that is rapidly growing. It covers the domain or subsets including machine

learning and deep learning inside it as shown in **Figure 4**, whose methodologies and algorithms can be used to solve real-world problems.

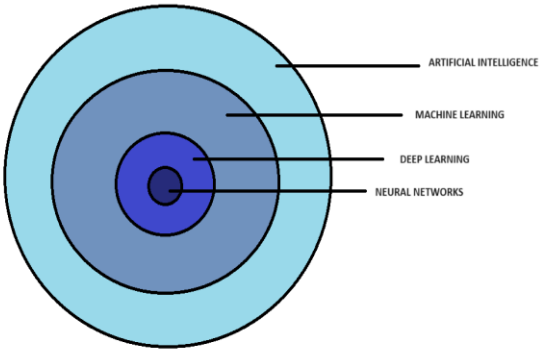

Figure 4. Domains of AIs

3.1. Neural Networks

AI is a vast topic which includes domains inside it such as Machine Learning, & one of the most efficient classes of algorithm of Machine Learning is Deep Learning. AI works on the principle of various different kinds of Neural Networks, including Artificial Neural Network (ANN), Dense Neural Network (DNN) [3], Multi-layer Perceptron, Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) [4]. Each neural network has its own architecture & applications along with limitations. The most feasible neural networks for cancer diagnostics are ANN, DNN & CNN. Due to the flexibility of neural networks, they are being implemented in many fields of biology, such as medicine [5], healthcare [6], drug discovery [7,8], medical imaging and radiology [9,10], etc.

The architecture of some of them are as follows:

3.1.1. Perceptron & Artificial Neural Network

Perceptron is a combination of layers of neurons composed of mathematical functions, connected together to process input to achieve desired output. The layers of perceptron include Input layer, Hidden Layer & Output Layer. Input layer is responsible for gathering data, preferably pre-processed for travelling through the other layers of Perceptron. Hidden layer has multiple neurons, which are responsible for receiving & processing data to progress the desired output. Output layer has ending neurons i.e. final desired classes of the whole perceptron, as shown in **Figure 5**. The layers of perceptron are interconnected to provide fully connected neural network, in which the mathematical interconnection is termed as "weights". The perceptron takes in a vector, multiply the weights & bias in it (for standardized values), so that the input passes as Net input. The net input is then passed into an activation function to map the input into desirable classes. Perceptron provides bi-directional data propagation, i.e. both forward & backward propagation, due to which the whole process may take time to analyse the solution [11]. On the contrary, ANN is a group of multiple perceptron or neurons at each layer, which offers only forward propagation, as shown in **Figure 6.** Therefore, they are also known as Feed-Forward Neural Networks [12]. Their functioning becomes more interpretable due to lesser neurons & complexity [13,14].

The functioning of the whole ANN majorly depends on the type of mathematical functions we incorporate in it. Two such functions are Loss Function & Activation Function. **Loss function (LF)** is a mathematical function that repeatedly estimates the error of the model, so that the error can be decreased by updating the weights associated with each layer. Some of the popular LF are Regression LF, Binary Classification LF, and Multi-class Classification LF. The techniques for minimizing the loss inside these loss functions include mean squared error loss, mean squared logarithmic error loss, mean absolute error loss (for Regression LF); binary cross entropy, hinge loss, squared hinge loss (for Binary Classification LF); and multi-class cross entropy, sparse multi-class cross entropy and Kullback-Leiblar divergence (for Multi-class LF). **Activation function** is a mathematical function that maps the input to be converted into output. It is highly dependent on the type & category of data & it is very crucial for determining the accuracy of the model as incompatible activation function may result in fall of accuracy of the model. Activation function is of many types based on the type of data. Some of the examples are linear, sigmoid, Rectified Linear Unit or ReLU & softmax activation function. The linear activation function maps the input linearly with output, therefore it's feasible for simple data. The sigmoid activation function maps the input to only binary values i.e. 0 & 1 based on the closeness of input data towards these digits. It is suitable for binary classification of data into 2 classes. The ReLU is considered to be one of the best activation function

due to its less computational requirements & less complexity. It maps the input values to either 0 (if negative) or that exact same value (if positive). Therefore, ReLU activation function is highly feasible for non-linear data. Finally, the softmax activation function maps the input values to probabilities or confidence score based on the similarity of input with particular output class. The classification is done on the basis of input that has highest probability related to particular output class. Therefore, it is suitable for multi-class classification.

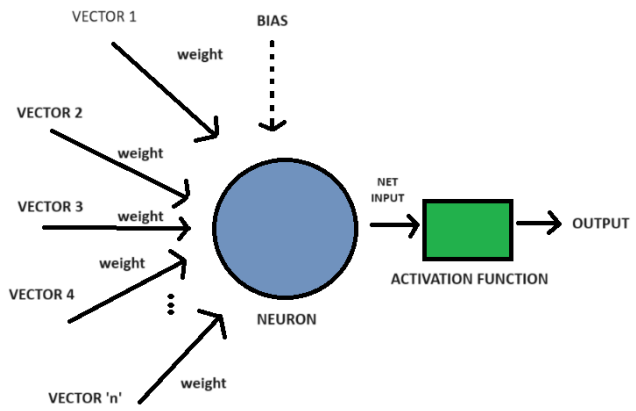

Figure 5. Architecture of Perceptron

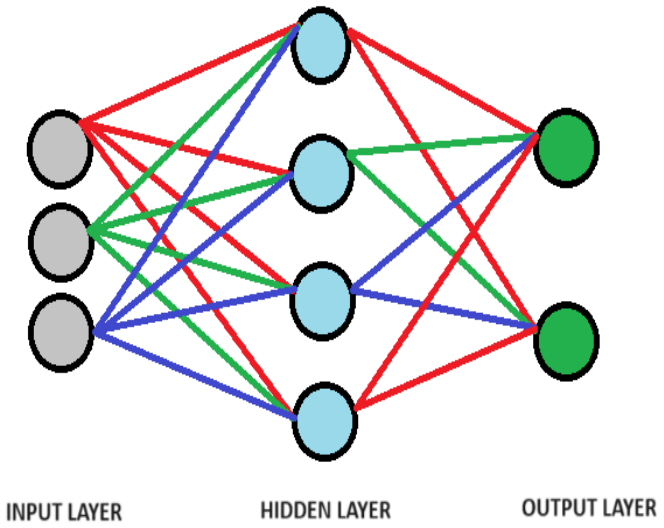

Figure 6. Architecture of Artificial Neural Network (ANN)

3.1.2. Convolutional Neural Network

A CNN is a special type of ANN which is mainly used for image & object classification due to its potential of recognizing patterns & details from an image or object. CNN can be implemented in disease diagnostics by complementing it with conventional medical imaging techniques such as radiology, MRI & CT, etc.

CNN is very popular for having an optimized architecture that can be customized as per need w.r.t. the application & the use-case [15–18]. CNN Architecture includes a set of layers, primarily in the pre-defined sequence: Input Layer, alternate Convolution Layer & Pooling Layer, Fully Connected Layer and the Output Layer, as shown in **Figure 7.** The **input layer** gathers data of an image having particular dimensions & channels based on colour i.e. 3 channels for 3 primary colours including Red, Green and Blue, respectively. The **Convolution layer** is the building block of whole CNN as it is responsible for doing most of the heavy computational task i.e. feature extraction. Convolution layer extracts features from an image based on pixels & channels by using matrices of different orders based on the weighted filter (mathematical function i.e. multiplied with input data to obtain desirable values, padding is done for optimizing the result). Filter is smaller in dimension as compared to input matrix, so that it's passed over different sections of input matrix to extract features. **Pooling layer** is responsible for dimensionality reduction to shorten the image for better results, by reducing complexity & parameters of input through weightless filters. Pooling layer may use two common technique for dimensionality reduction, which are max-pooling & average pooling. In maximum pooling, the maximum value in input matrix is considered to be passes through the filter to form new compact matrix whereas in Average pooling, the average value of input matrix is passed into filter to form new compact matrix, , as shown in **Figure 8. Fully Connected Layer** flattens the processed matrix into a vector so that it can passed into a fully connected Neural Network to obtain the desirable output based on activation function. **Output layer** consists the classes or categories of our desired result. Activation function returns final result in terms of numerical value such as probability, binary digits, etc. based on which the input data is classified to particular output class [19].

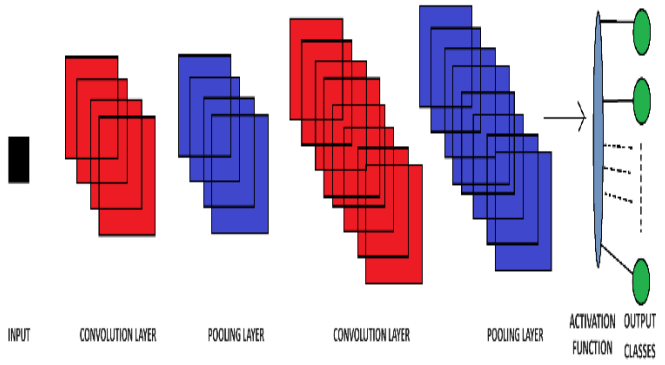

Figure 7. Architecture of Convolutional Neural Networks (CNN)

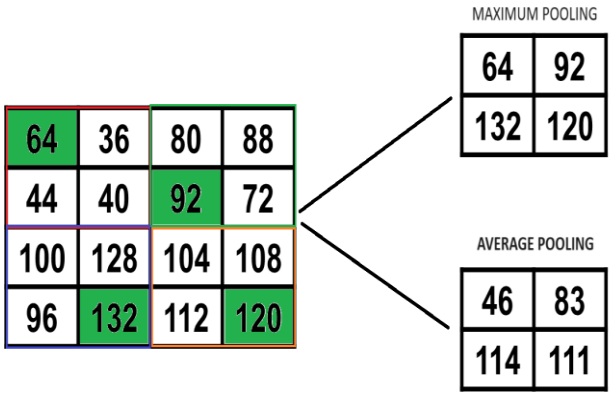

Figure 8. Pooling techniques in Convolutional Neural Networks (CNN)

3.2. Optimization techniques for AI Models and Neural Networks

Sometimes, AI models may fail to perform well on test data even though they show very high accuracy during evaluation. This may occur due to some conditions such as over-fitting, under-fitting, etc. Overfitting makes AI model to perform well for train data but very poor for test data whereas in under-fitting, AI model performs poor for both train & test data. These problems can be solved by optimizing the model & the neural network by adjusting weights & bias continuously as per need, to reduce the error in output. Some of the techniques are as follows:

3.2.1. Gradient Descent (GD)

Gradient descent is an iterative optimization algorithm that is used to find local minimum/maximum of the mathematical function of error/accuracy (typically the LF) to decide other parameters of model for better optimization. Based on the data, we can implement different types of GD, such as, Batch GD, Mini-Batch GD, and Stochastic GD.

In Batch GD, all the samples of the main dataset are processed in one training process. Therefore, it's computationally expensive & time consuming. On the contrary, Mini-Batch GD is very feasible for such computationally heavy applications as mini-batches or samples of dataset are processed instead of whole dataset, which makes it faster & computationally light to implement. In Stochastic GD, each data is trained individually, therefore number of iterations is very large in it.

3.2.2. Back Propagation (BP)

Backpropagation is an optimization technique that uses the concept of GD. The BP algorithm looks for the local minimum of error, i.e., lowest value of error through GD. Based on the local minimum, BP algorithm iterates back over the model's weights to adjust them again & again until the optimum weight is obtained having lowest value of error. The weights that gives lowest value of error are considered to be the solution for the optimization.

3.2.3. Batch Normalization

Normalization is the mathematical process of standardizing a data & mapping it to a value that lies in a fixed range, i.e., -1 to +1 or 0 to +1. Normalization is applied at data to make it uniform. Similarly, in neural networks, Batch Normalization is applied to input layer to standardize the inputs for better processing into the neural network. It aims to make the mean to be 0 & standard deviation to be 1 so that the data behaves as standard normal distribution.

3.2.4. Optimizers

Optimizers are mathematical functions that can be implemented to a neural network to increase the accuracy. Some of the famous optimizers include Nesterov Accelerated Gradient or NAG, Adaptive Gradient Algorithm or AdaGrad, AdaDelta, Root Mean Square Propagation or RMSProp and Adaptive Momentum Estimation or AdaM. Any one of these can be implemented in the neural network based on

compatibility & requirement of the model as well as the application.

4. Cancer Staging through TNM Method

The stage of cancer is very crucial for selecting the treatment of patient as different types of treatment procedures are followed for different stages of cancer. Therefore, determination of stage of cancer, i.e., staging is very crucial for treatment process. The whole diagnostic and treatment procedure of cancer is done by a team of medical specialists, i.e., known as multi-disciplinary team (MDT). MDT includes physician, radiologist, histo-pathologist, and oncologist. The workflow of cancer treatment starts from initial tests and biopsy by physicians, followed by tissue analysis by histo-pathologist and imaging techniques and analysis to determine staging by Radiologists. Once the stage is confirmed, the treatment is determined based on the stage and gets started immediately as per suggestions of oncologist. Clinical oncologists and surgeons treats initial stages of cancer through radiation therapies, and surgery while medical oncologists are specialized for higher treatment including Chemotherapy and Hormone therapy.

One of the most famous procedures for determining the stage of cancer is TNM procedure, i.e., Tumour, Node, and Metastasis, which was developed by Professor Pierre Denoix in the duration of 1943-1952. TNM is based on the size and count of Tumours (zero tumour or T0, one tumour or T1, and so on), spread of cancer to local nodes and lymphatic system (zero node or N0, one node or N1, and so on), and the movement of tumour, or Metastasis (absence of metastasis or M0, and presence of metastasis or M1). Based on the TNM criteria, cancer may have four stages, namely stage 1, stage 2, stage 3 and stage 4, respectively. Stage 1 and 2 are contained within organ but initial metastasis starts from stage 3, which is fatal if not diagnosed earlier. Stage 1, 2, 3 and 4 are characterised by TNM features as (T1, N0, M0), (T2-3, N0-1, M0), (T4, N1-2, M4) and (T>4, N>4, M1), respectively. Therefore it is very crucial to determine stage of cancer at an early stage to save the life of patient. In this era, AI is being developed and implemented in cancer diagnostics to determine the staging and support the treatment efficiently [20].

5. Traditional Biological Methods & AI Methods for Cancer Diagnostics

5.1. Brain Cancer

Brain cancer or tumour is caused by change in the genetic material or DNA of cancer cells which is either triggered by inherent genetic composition or through environment. Brain tumour can be of many types, primarily malignant & benign, based on their locomotion. Majority of the brain tumours comes under Glioma family, out of which Astrocytoma is the deadliest brain tumour. Detection of such tumours is very time consuming through conventional, methods therefore, many researchers implemented AI to detect brain tumour [21].

Joshi *et al.* [22] developed a neuro-fuzzy classifier to detect brain tumour, specifically an ANN based on MRI images to detect Astrocytoma brain tumour in patients. The workflow of ANN includes image processing, feature extraction & image classification. Image processing involved many techniques including image segmentation, histogram equalization, etc. in order to achieve accurate results. Feature extraction used Gray level Co-Occurrence Matrix (GLCM) & other thresholding techniques. Finally, an ANN was developed, which was trained & tested on MRI images of patients diagnosed with Astrocytoma type of brain tumour. Results of ANN were complementary with the known Knowledge base & were accurate. The model could be improved by using better images of other types (PET, MRS, CTS, etc.).

5.2. Lungs Cancer

Lungs cancer is one of most fatal type of cancer, which is caused due to abnormal growth of cells in lungs. This type of cancer is accelerated through smoking & environment, which harms the lungs. Most of the cases of lungs cancer are diagnosed at middle or advanced stage. Therefore, many researchers worked on using AI for lungs cancer detection.

R. Bellotti *et al.* [23] developed a CAD system for detection of lungs cancer in CT images. The accuracy obtained was 88.5%. Van Ginnekan *et al.* [24] combined six CAD algorithms to detect cancer

nodules. The accuracy obtained was 80%. Al-Kadi *et al.* [25] developed a system based on fractal texture features. The accuracy was 83.3%. Zakaria Suliman Zubi *et al.* [26] developed an ANN CAD system for lungs cancer detection, based on the parameters such as area, perimeter & shape of tumour. The accuracy obtained was 90%. Jinsa Kuruvilla *et al.* [27] developed a neural network for detection of lungs cancer, based on the statistical parameters including mean, standard deviation, skewness, kurtosis, fifth central moment & sixth central moment. The accuracy obtained with Traingdx (GD with variable learning rate & momentum) obtained was 91.1%. They have also proposed two training functions with better performance, i.e., 1st training function gives 93.3% accuracy, 100% specificity and 91.4% sensitivity, whereas, 2nd training function gives 93.3% accuracy & 0.0942 mean square error. Fatma Taher *et al.* [28] developed an ANN with Fuzzy clustering method to detect lungs cancer. They used high-grade segmentation & thresholding algorithms for obtaining high accuracy as gray level & contrast in images complicates the detection process. They have used 2 segmentation methods, including Hopfield Neural Network (HNN) and a Fuzzy C-Mean (FCM) clustering algorithm for better segmentation. Their model proved to be a better alternative as compared to manual method of analysis of sputum samples.

5.3. Breast Cancer

Breast cancer is development of tumour by abnormally growing cells inside milk ducts or lobules of women. Breast cancer is one of the major causes of women cancer mortality. There are four major types of breast cancer including normal, benign, in-situ carcinoma and invasive carcinoma, out of which invasive carcinoma is most fatal. Conventional histopathological imaging techniques for breast cancer detection involves X-ray mammography, Magnetic Resonance Imaging (MRI), Computed Tomography (CT), Ultrasound (US), Portion Emission Tomography (PET) and temperature measurement, out of which Mammography is considered to be one of the most efficient ways for breast cancer detection. Breast cancer can also be diagnosed by radio genomics comprising medical imaging and gene expression inside it. The common parameters for breast cancer detection includes tumour size, number of metastasis record, size & density of epithelial cells, clump thickness, symmetry & uniformity of cell size and compactness. Also, radiologists examine size, shape, number and distribution of microcalcifications in breast lesions or carcinoma through mammography to detect breast cancer in conventional way. For the better communication and radiologists for breast cancer detection, the American College of Radiology (ACR) developed a reporting system known as Breast Imaging Report and Data System (Bi-RADS). BI-RADS considered breast density too as a parameter to determine breast cancer. In order to achieve early diagnosis of breast cancer, many researchers tried to incorporate AI into it. Some examples are as follows:

Turgay Ayer *et al.* [29] developed an ANN BP algorithm and early-stopping technique to detect breast cancer. The parameters considered include mammographic descriptors, demographic factors & BI-RADS categories. The specificity obtained in ROC curve by 10-fold cross validation technique was around 90%. Nan Wu *et al.* [30] developed a CNN based on the features including mammographic phenotype, pixel-labels, and BI-RADs. The model is based on ResNet that was trained on 229,426 digital mammography images from 141,473 patients. The accuracy in Area Under the Curve (AUC) graph was found to be 89.5%. D. Sun *et al.* [31] developed a deep neural network by integrating multi-dimensional data for breast cancer detection. Park *et al.* [32] proposed a semi-supervised learning model for breast cancer detection. Jhajharia *et al.* [33] developed an ANN with Principal Component Analysis algorithm for feature extraction in order to detect breast cancer. Xu *et al.* [34] developed a support vector machine based recursive feature examination Support Vector Machine (SVM-RFE) model for gene selection and breast cancer detection. Delen *et al.* [35] developed an ANN with decision tree algorithms and logistic regression (LR) for breast cancer detection. The accuracy was found to be 93.6% with decision tree (C5). Yuanjiao *et al.* [36] developed a computerized microcalcification detection system based on fuzzy logic, vibroacoustography and probabilistic neural networks on mammograms data along with machine learning algorithms such as SVM, LDA, NDA, and PCA. Shlomi laufer *et al.* [37] developed a system of self-organizing map with weight adjustment for breast cancer detection. Xiangchun Xiong *et al.* [38] developed a probabilistic neural network based on statistical methods and data mining methods including linear regression, PCA, and PLS. David B.

Fogel *et al.* [39] developed an ANN with Linear Discriminant and Principal Component Analysis methods for breast cancer detection. R.R. Janghel *et al.* [40] developed a Multi-Layer Perceptron (MLP) Neural networks along with optimization methods including BP, Radial Basis Function or RBF and LVQ. The training of the model is based on Wisconsin Breast Cancer Diagnosis (WBCD) database. The evaluation using Mean Sum-squared Error (MSE) evaluated the accuracy of MLP, RBF & Linear Vector Quantization or LVQ to be 51.88%, 49.79% and 95.82%, respectively. Therefore, MLP with LVQ method is the best alternative for their model. Kowal *et al.* [41] developed an AI model based on K-nearest neighbours, Naïve Bayes (NB) and Decision tree algorithms for breast cancer detection. The model did image segmentation for nuclei region and passing segmented region for image classification based on these algorithms, and was successful in determining breast cancer based on images dataset. Filipczuk *et al.* [41] proposed a CAD system for breast cancer detection. The approach involved image segmentation of nuclei regions through circular Hough transform, and passing segmented regions to image classifiers based on K-nearest neighbours, NB, SVM and Decision tree algorithms. The accuracy obtained was found to be 98.51%. Zhang *et al.* [42] developed an AI model based on SVM and MLP methods. The model used local binary pattern technique for feature extraction from biopsy test images, so that the input data can be fed to the model. The accuracy obtained was found to be 97%. Sertan Kaymak *et al.* [43] developed an AI model for breast cancer detection. The model was based neural networks involving BP and RBF methods for better results. The model was trained on the dataset of 176 digital images. The model used image pre-processing techniques for better accuracy, some of which include gray scale conversion, Gaussian filter, and Haar discrete wavelets calculation. The evaluation revealed mean squared error and accuracy of model trained on BP and RBF to be 0.01352, 29.02% and 0.0220, 70.49%, respectively. Jeffrey Alan Golden *et al.* [44] proposed nearly 32 methods for breast cancer detection, where most of them are based on CNN. The methods were presented in CAMELYON16 i.e. Cancer Metastasis in Lymph Nodes Challenge contest. The evaluation revealed the accuracy by Area under the Curve (AUC) method for the best model to be 94.3%. Ismail Saritas *et al.* [45] developed an ANN for breast cancer detection based on the parameters including Bi-RADS report, age, shape, density, margin, bias and thickness of breast lesion.

5.4. Skin Cancer

Skin cancer is the rapid birth & growth of skin cells or melanocytes, which may occur due to several factors including genetic factors or through environmental stimulation such as exposure to ultraviolet radiations. Out of all types of skin cancer, melanoma is the deadliest as 75% deaths by skin cancer occur due to Melanoma. During melanoma, additional cells are created & old cells die unnecessarily, which eventually leads to aggregation of the extra cells to form a tumour. The skin cancer diagnosis is a cumbersome process, even for a specialist as symptoms appear when the cancer has crossed its initial stages. The symptoms may include occurrence of dark pigmentation on different areas of skin, which usually shows black or brown colouration. In order to detect skin cancer in earlier stages, many researchers have tried to incorporate AI into it. Some of them are:

Gao *et al.* [46] developed an image segmentation model for melanoma based on Delaunay triangulation method. Shivangi S Jain *et al.* [47] developed an ANN for skin cancer detection. They have used ABCD Rule, i.e., Asymmetry, Border, Colour, Diameter as parameters for skin cancer detection. The workflow included image segmentation and image classification based on ANN. The accuracy obtained was around 97%. Heller *et al.* [48] developed a CAD system for skin cancer. Esteva *et al.* [49] developed a CNN for skin cancer detection, at a dermatologist-level. Anas *et al.* [50] developed a CAD system for melanoma classification in 4 categories. Abbas *et al.* [51] developed skin cancer classification system using AdaBoost MC. Capdehourat *et al* [52]. developed skin cancer classification system using AdaBoost MC. Almansour *et al.* [53] developed skin cancer classification system using K-means Clustering and SVM classifier. Hitoshi *et al.* [54] proposed an automatic system for melanoma classification. Isasi *et al.* [55] proposed an automatic system for melanoma classification. Giotis *et al.* and Ruiz *et al.* [56] proposed decision support systems based on neural networks trained on the parameters including lesion texture & colour, affected area & location, phenotypic attributes, etc. Ulzii-Orshikh Dorj *et al.*

[57] developed a CAD system for skin cancer detection, based on pre-trained AlexNet CNN model for feature extraction and ECOC SVM classifier for image classification. The CNN model included 5 Convolution layers, 3 Maximum Pooling Layers, and 3 Fully Connected Layers. They have trained their model on 3753 images dataset for four types of skin cancer, namely, Actinic Keratoses (AJK), Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC), and Melanoma. The accuracy, sensitivity, and specificity was found to lie in the range of 90-99%, which shows the model is efficient enough to perform skin cancer detection. Mahamudal Hasan *et al.* [58] developed a CNN for skin cancer detection. The CNN successfully classified the skin cancer into Benign or Malign Melanoma. The accuracy obtained was around 80-90%. Ni Zhang *et al.* [59] developed a semi-supervised CNN based on some deep learning methodologies such as AlexNet, VGG-16, LIN, Inception-v3, and ResNet. They have used Dermquest and DemIS Digital Database for building the CNN model. The training parameters include ReLU activation function, 28000 epochs, and 45 training steps, respectively. The evaluation parameters include sensitivity, specificity, accuracy, NPV & PPV, respectively. They have use Matlab R2016 Software for building whole model.

5.5. Prostate Cancer

Prostate cancer is the abnormal growth of cells in prostate gland in males. It is usually caused due to age, genetic history and ethnic background. The conventional diagnostic techniques include genomic biomarkers, Gleason grading, transrectal ultrasonography, rectal examination and biopsy to check the amount of prostate-specific antigen (PSA) in blood. PSA levels above 10 ng/mL are considered to be an identifier of prostate cancer, but solely PSA levels can't guarantee the presence of prostate cancer. Therefore, other diagnostic techniques needs to be done in order to confirm prostate cancer. Prostate cancer detection requires many prostate biopsies due to which time and complexity of detection increases [60]. Therefore, early detection of prostate cancer is crucial for treatment, due to which AI algorithms can be developed to achieve this task. AI models also provide non-invasive and safer approach to detect prostate cancer.

Ismail Saritas *et al.* [61] developed an ANN or fuzzy expert system based on the parameters including patient's age, ethnic background, genetic history, PSA levels (tPSA, fPSA), Prostate volume and non-linear relationships among them. Djavan *et al.* [62] developed an ANN for determining the PSA levels based on sensitivity and specificity of the model. The Receiver Operating Characteristic Curve was found to be a major parameter while building the model. Finne *et al.* [63] developed an AI model based on the parameters including data on total PSA, free PSA, Digital Rectal Examination (DRE) and prostate volume from serum samples of 656 men. The specificity obtained was found to be 95% which suggested that ANN could reduce false-positive results. Ge *et al.* [64] developed an ANN & LR algorithm to detect prostate cancer. The parameters considered while training include free PSA, prostate volume, age and prostate density from 586 men diagnosed with cancer i.e. confirmed by biopsy results. Stephan *et al.* [65] developed an ANN based on the parameters including free PSA %, total PSA, patient's age, prostate volume and digital rectal examination data. The model was successful in decreasing occurrence of false positive results. Strom *et al.* [66] developed an ANN based on digital histopathological slide images from 1247 men to predict the Gleason grade of malignant tumour. The accuracy obtained is 99.7% (area under ROC), which helped to assess biopsies. Lawrentschuk *et al.* [67] developed an ANN based on polychotomous LR model to predict biopsy result of 3025 men. The parameters considered are age, PSA levels, digital rectal examination data, prostate volume, and positive trans-rectal ultrasound (TRUS) data. Remzi *et al.* [68] developed an ANN based on the parameters including age, PSA levels, free to total PSA level ratio, PSA velocity, prostate volume, and data from digital rectal examination & trans-rectal ultrasound from 820 men. The specificity and sensitivity of the model were found to be 38% and 95%, respectively. Fehr *et al.* [69] developed a machine learning based classifier algorithm for prostate cancer detection. This model considered MRI images based texture features for determining the level of Gleason grades. The accuracy obtained was around 93%. Antonelli *et al.* [70] developed a machine learning based classifier algorithm for prostate cancer detection. It predicted Gleason grades in prostate tumor efficiently. Toivonen *et al.* [71] developed an AI model based on the parameters including T2-weighted high quality MRI images, Diffusion weighted Imaging for texture features, and

T2 relaxation maps from 100 patients. The model performed well in accordance with Gleason scores.

5.6. Ovarian Cancer

Ovarian cancer is the abnormal growth of ovarian cells, especially due to cells outside ovary which causes epithelial type ovarian cancer. The cause may be genetic, age, irregular initiation & termination of menstruation, and hormones level. Ovarian cancer is fatal gynaecological cancer for women as it is fifth most common cause for deaths due to cancer in women. Due to such a fatal threat, it is necessary to detect ovarian cancer at an early stage to treat it effectively. Therefore, many researchers have tried to incorporate AI for early cancer diagnostics. Some of them are as follows:

Munetoshi Akazawa *et al.* [72] proposed a system for ovarian cancer detection using AI model based on Machine Learning and Neural Networks. The model used five classifier algorithms, namely LR, NB, Random Forest or RF, SVM and XGBoost or XGB. The model was based on 16 parameters, including age, gravidity, parity, menopause, endometriosis, body mass index, white blood cell or WBC count, haemoglobin or Hb, platelet count, albumin, C-reactive protein or CRP, carcinoembryonic antigen, size of tumour, ascites. Out of these 16 features, CRP and ascites were found to be best predictive features. The accuracy obtained for XGB, RF, LR, and SVM algorithms are 80%, 78%, 67%, and 62%, respectively.

5.7. Cervical Cancer

Cervical cancer is the 2nd most common cancer in women. Cervical cancer is not diagnosed at early stages due to complexity in conventional diagnostic techniques, mainly Pap smear screening & Liquid Cytology Based (LCB) screening. In Pap smear screening, cervical cells are manually collected from uterus through a brush or spatula and are analysed in the laboratory to manually classify normal and cancerous cells. This method is vulnerable to false results due to human errors, and it is monetarily expensive. In Liquid Cytology Based (LCB) screening, the cervical cells are manually collected from uterus and are analysed in laboratory with chemicals such as acetic acid. The presence of particular colouration in the sample signifies the presence of cancerous cells in it. These traditional methods are time consuming and are expensive therefore there is a need to develop an effective non-invasive method that can detect cervical cancer at an early stage. Many researchers have tried to incorporate AI for quicker diagnosis, and were successful in it. Some methods are as follows:

M. Anousouya Devi *et al.* [73] proposed an ANN or feed-forward neural network to detect cervical cancer. Endah Purwanti *et al.* [74] developed an ANN along with Learning Vector Quantification technique for better results in cervical cancer detection. The accuracy obtained was found to be 90%. N. Wickramsinghe *et al.* [75] proposed an ANN consisting MLP for cervical cancer detection. Aabha S. Phatak *et al.* [76] developed an ANN with SVM algorithm for cervical cancer detection. Praba *et al.* [77] developed a neural network with K-nearest neighbours & Bayes classifiers for cervical cancer detection. This model was working well with agreement to the Pap smear screening method. N. Mustafa *et al.* [78] developed an ANN based on Pap smear screening test images. The test parameters include colour intensity (RGB), perimeter and area, etc. Babak Sokouti *et al.* [79] developed a Levenberg-Marquarat feed forward MLP neural network for cervical cancer detection. Isabelle Claude *et al.* [80] developed an ANN for cervical cancer detection. The model focussed on spatial parameters and contour detection to classify cancer cells. Ganesan *et al.* [81] developed an ANN based on dimorphic data. They have incorporated BP method to adjust weights of the ANN model to increase the accuracy. Paulo J. Lisboa *et al.* [82] developed an ANN for cervical cancer detection for enhancing the decision support systems. Intan Aidha Yusoff *et al.* [83] developed a neural network (Cascaded Multi-layer Perceptron, C-MLP) which gave input as cell features to be processed in the network. Zati Athiar Ramli *et al.* [84] developed a hybrid Multi-Layer Perceptron (H-MLP) with least square algorithm, as a non-linear model to detect cervical cancer. Mehdi Bazoon *et al.* [85] developed an ANN with BPr and ART for cervical cancer detection. Kagan Tumer *et al.* [86] developed a feed forward neural network or MLP based on RBF and ensemble algorithms. Francisco J. Gallegos-Funes *et al.* [87] developed a neural network based on RBF to extract features of cytoplasm and nucleus for cervical cancer detection. Kwang Balk Kim *et al.* [88] developed a Fuzzy RBF network based on image segmentation of nucleus from cell and classification through K-Means Clustering algorithm

based on RGB values. Bondarenko *et al.* [89] developed a neural network, combined with self-organization maps for distinguishing abnormal cells from normal cells. Nikolaos Ampazis *et al.* [90] proposed a new 2nd order NN based on LMAM and OLMAM (Optimized – Levenberg-Marquardt with Adaptive Momentum) algorithms for training, which are usually used for non-linear data.

The summary of cancer diagnostic techniques using methods AI can be seen in **Table 1** and the common parameters that are usually used to train an AI model for cancer diagnostic application can be seen in **Table 2**, respectively.

Table 1. Summary of the contributions of authors for cancer diagnostics

S. No.	Type of cancer	AI method	Reference
1.	Brain cancer	• Neuro Fuzzy ANN classifier	• Joshi Jain *et al.* [22]
2.	Lungs cancer	• CAD system • CAD system based on 6 algorithms • ANN CAD system • ANN with GD • ANN with fuzzy clustering	• R. Bellotti *et al.* [23] • Van Ginnekan *et al.* [24] • Zakaria Suliman Zubi *et al.* [26] • Jinsa Kuruvilla *et al.* [27] • Fatma Taher *et al.* [28]
3.	Breast cancer	• ANN with BP and early-stopping • CNN • DNN • Semi-supervised ML model • ANN with PCA • SVM with RFE • ANN with Decision tree and Logistic regression • Fuzzy NN with SVM, LDA, NDA and PCA • Self-organizing maps • Probabilistic NN with PCA and PLS • ANN with LD and PCA • MLP with BP, RBF and LVQ • CAD with K-NN, NB and DT algorithms • CAD system with K-NN, NB, DT and SVM • MLP with SVM • NN with BP and RBF • CNN • ANN	• Turgay Ayer *et al.* [29] • Nan Wu *et al.* [30] • D. Sun *et al.* [31] • Park *et al.* [32] • Jhajharia *et al.* [33] • Xu *et al.* [34] • Delen *et al.* [35] • Yuanjiao *et al.* [36] • Shlomi laufer *et al.* [37] • Xiangchun Xiong *et al.* [38] • David B. Fogel *et al.* [39] • R.R. Janghel *et al.* [40] • Kowal *et al.* [41] • Filipczuk *et al.* [41] • Zhang *et al.* [42] • Sertan Kaymak *et al.* [43] • Jeffrey Alan Golden *et al.* [44] • Ismail Saritas *et al.* [45]
4.	Skin Cancer	• NN with Delaunay triangulation method. • ANN with ABCD rule • CAD system • Dermatologist-level CNN • CAD system • AdaBoost MC algorithm • AdaBoost MC algorithm • NN with K-Means Clustering & SVM • CAD system • CAD system • Decision support systems • CNN with AlexNet and ECOC SVM Classifier • CNN • Semi-Supervised CNN	• Gao *et al.* [46] • Shivangi S Jain *et al.* [47] • Heller *et al.* [48] • Esteva *et al.* [49] • Anas *et al.* [50] • Abbas *et al.* [51] • Capdehourat *et al.* [52] • Almansour *et al.* [53] • Hitoshi *et al.* [54] • Isasi *et al.* [55] • Giotis *et al.* and Ruiz *et al.* [56] • Ulzii-Orshikh Dorj *et al.* [57] • Mahamudal Hasan *et al.* [58] • Ni Zhang *et al.* [59]
5.	Prostate cancer	• ANN fuzzy expert system • ANN • CAD system • ANN with logistic regression • ANN • ANN • ANN with logistic regression • ANN	• Ismail Saritas *et al.* [61] • Djavan *et al.* [62] • Finne *et al.* [63] • Ge *et al.* [64] • Stephan *et al.* [65] • Strom *et al.* [66] • Lawrentschuk *et al.* [67] • Remzi *et al.* [68]

		• ML classifier • ML classifier • CAD system	• Fehr *et al.* [69] • Antonelli *et al.* [70] • Toivonen *et al.* [71]
6.	Ovarian Cancer	• CAD system with logistic regression, Naive Bayes, Random Forest, SVM and XGB algorithms	• Munetoshi Akazawa *et al.* [72]
7.	Cervical Cancer	• ANN • ANN with LVQ • ANN with MLP • ANN with SVM • NN with KNN and NB Classifier • ANN • Levenberg-Marquarat feed forward MLP • ANN • ANN • ANN • Cascaded MLP • Hybrid MLP • ANN with BP and ART • MLP and RBF • RBF • Fuzzy network with RBF • NN • NN with LMAM and OLMAM	• M. Anousouya Devi *et al.* [73] • Endah Purwanti *et al.* [74] • N. Wickramsinghe *et al.* [75] • Aabha S. Phatak *et al.* [76] • Praba *et al.* [77] • N. Mustafa *et al* [78]. • Babak Sokouti *et al.* [79] • Isabelle Claude *et al.* [80] • Ganesan *et al.* [81] • Paulo J. Lisboa *et al.* [82] • Intan Aidha Yusoff *et al.* [83] • Zati Athiar Ramli *et al.* [84] • Mehdi Bazoon *et al.* [85] • Kagan Tumer *et al.* [86] • F. J. Gallegos-Funes *et al.* [87] • Kwang Balk Kim *et al.* [88] • Bondarenko *et al.* [89] • Nikolaos Ampazis *et al.* [90]

Table 2. Common training parameters used in AI models

S. No.	Type of cancer	Common parameters used in AI models
1.	Brain cancer	Tumour size, RGB intensity.
2.	Lungs cancer	Area, perimeter, shape and size of tumour.
3.	Breast cancer	Tumour size, metastasis, clump thickness, density, margin, bias, mammographic descriptorsm, demographic factors, Patient's age, and BI-RADS report. Epithelial cells size, shape, symmetry and uniformity, number and description of micro-calcifications.
4.	Skin Cancer	Asymmetry, Border, Colour, Diameter of tumour, lesion texture & colour, affected area & location, phenotypic attributes.
5.	Prostate cancer	Patient's age, ethnic background, genetic history, PSA levels (total or tPSA, free or fPSA), Prostate volume, Digital Rectal Examination and trans-rectal ultrasound (TRUS) data.
6.	Ovarian Cancer	Patient's age, gravidity, parity, menopause, endometriosis, body mass index, WBC count, Hb, platelet count, albumin, CRP, carcinoembryonic antigen, size of tumour, ascites.
7.	Cervical Cancer	Colour intensity (RGB), perimeter and area, spatial parameters, contour detection.

6. Future Aspects of AI and Industry 4.0

AI is a rapidly growing field. AI may be classidfed into three types, namely Weak or Narrow AI, General or Strong AI and Super AI. Currently, the type of AI which is being developed is weak or narrow AI, whose applications remain constricted to specific task only. There are numerous hypothesis and theories regarding the development of AI which may be intelligent enough as human (Artificial General Intelligence or AGI) or may even surpass human intelligence (Artificial Super Intelligence or ASI). Due to the flexibility of AI, it will be implemented in most of the applications in future. Some of the fascinating applications of AI include:

6.1. AI in Robotics

In the recent times, AI has entered into the domain of robotics. For example: Human Follower Robot is based on the use of AI algorithm to detect human based on Machine learning and Deep learning algorithms. The robot may have a camera module attached to the body for live implementation of robotics.

6.2. AI in Agriculture

AI can be used in agriculture to enhance the productivity for meeting the demands of great populations. For example: Machine Learning algorithms can be implemented sensor data (temperature, humidity, power fluctuations, etc.) from agricultural field to detect fault in setup.

6.3. AI in Education

AI can be used in education system to enhance the education of students by providing quicker solutions. For example: Machine learning and Deep learning algorithms may be used with a camera module for facial recognition to build a virtual attendance system for saving time.

6.4. Digital Twin Technology

Digital twin technology is a multi-disciplinary simulation technique that uses the knowledge of simulation, augmented reality and virtual reality for mapping the physical reality into virtual workspace. It is a three-dimensional modelling technique or a Cyber-Physical system to build real-time digital copies of physical objects [91]. It is a developing technology of Industry 4.0 which will be in demand in future. The Digital Twin technology was introduced by Michael Grieves in 2003 at University of Michigan. Digital twin technology was called mirror-spaced model or information mirror model in the early stages of development. National Science Foundation of US introduced Cyber-Physical System (CPS) in 2006 based on digital twin technology whereas NASA publicized this technology in 2011, respectively.

The mechanical and technological workflow involves the data interfacing or integration between physical and virtual objects. Digital twin technology is multi-physical, dynamic and super-realistic tool that involves computer numerical control (CNC) algorithms with probabilistic approaches, known as digital thread, to replicate physical objects into virtual workspace. The concept of digital thread can be used for digital manufacturing to aid the manufacturing process in industries. Digital twin technology involves the usage of design tools such as voxel-based discretization techniques, geometric multi-rid solvers and hardware-aware implementations for building an application [92].

Some applications of digital twin technology includes product design, workshop design & production scheduling optimization, process and production planning and control, product quality analysis, product's error monitoring, robot milling, additive manufacturing and intelligent sensor systems.

7. Conclusion

Cancer is fatal and one of the deadliest diseases, as it's incurable at higher stages. Therefore, a need to develop faster diagnostic system for cancer diagnostics led to the implementation of AI for medical diagnostics. As AI (AI) has a great potential to manage huge data, and to develop human-like computer models based on machine learning and deep learning algorithms, therefore it is used in various sectors including medical diagnostics, digital twin technology, etc. In this paper, we have listed many methods for diagnostics of different types of cancer. The most common techniques for cancer detection is found to be Neural Networks, especially ANN and CNN. Many machine learning algorithms and deep learning techniques were also used with neural network to minimize the loss for better and accurate results. For building an optimized AI solution to any problem statement, model's training parameters, hyper-parameter tuning and optimization techniques play a crucial role. Based on the current understanding and statistics, AI will become more accurate in the upcoming details, which will eventually help to solve real-life problems, if yielded properly. Thereby, AI will aid early cancer diagnostics with proper and accurate results through the CAD applications.

Abbreviations used

List of Abbreviations	
LF	Loss Function
ReLU	Rectified Linear Unit
BP	Back propagation
RBF	Radial basis function
LVQ	Linear Vector Quantization
CAD	Computer-Aided Detection
NAG	Nesterov Accelerated Gradient
AdaGrad	Adaptive Gradient Algorithm
RMSProp	Root Mean Square Propagation
AdaM	Adaptive Momentum Estimation
GD	Gradient Descent
MLP	Multi-Layer Perceptron
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
ART	Adaptive Resonance Theory
LR	Logistic regression
KNN	K-Nearest Neighbour

NB	Naïve Bayes
RF	Random Forest
SVM	Support Vector Machine
XGB	XGBoost
WBC	White Blood Cell
Hb	Haemoglobin
CRP	C-Reactive Protein
PSA	Prostate-Specific Antigen

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